# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 73.1 ± 4.72 |
| Correlation Stability | 83.6 ± 5.35 |
| Distribution Consistency | 82.1 ± 6.58 |
| Entropy Preservation | 80.8 ± 2.18 |
| Mean Z Score Drift | 0.3 ± 0.14 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 21.2 ± 4.31 |
| Data Density | 98.1 ± 0.84 |
| Coverage Score | 64.1 ± 11.67 |
| Feature Accuracy | 83.2 ± 7.74 |
| Novelty Score | 0.6 ± 0.10 |

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## Dataset Audit: parkinsons.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 195 |
| Columns | 23 |
| Numeric features | 23 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4o-mini |
| Folds (*13) | 10 |
| Context % (*14) | 20% |
| Generation % (*15) | 100% |
| Join context (*16) | True |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| MDVP:Fo | 154.229 | 154.033 | -0.196 | 0.1% | OK |
| MDVP:Fhi | 197.105 | 179.554 | -17.551 | 8.9% | OK |
| MDVP:Flo | 116.325 | 132.518 | +16.193 | 13.9% | Warn |
| MDVP:Jitter | 0.006 | 0.007 | +0.000 | 7.3% | OK |
| MDVP:Jitter.1 | 0.006 | 0.007 | +0.000 | 7.3% | OK |
| MDVP:RAP | 0.003 | 0.004 | +0.000 | 11.2% | Warn |
| MDVP:PPQ | 0.003 | 0.004 | +0.000 | 13.6% | Warn |
| Jitter:DDP | 0.010 | 0.010 | -0.000 | 0.6% | OK |
| MDVP:Shimmer | 0.030 | 0.031 | +0.001 | 2.9% | OK |
| MDVP:Shimmer.1 | 0.030 | 0.031 | +0.001 | 2.9% | OK |
| Shimmer:APQ3 | 0.016 | 0.017 | +0.001 | 7.1% | OK |
| Shimmer:APQ5 | 0.018 | 0.020 | +0.002 | 9.9% | OK |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 73.1 ± 4.72 |
| Correlation Stability (*2) | 83.6 ± 5.35 |
| Distribution Consistency (*3) | 82.1 ± 6.58 |
| Entropy Preservation (*4) | 80.8 ± 2.18 |
| Mean Z Score Drift (*5) | 0.3 ± 0.14 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 21.2 ± 4.31 |

| | |
|---|---|
| Data Density (*8) | 98.1 ± 0.84 |
| Coverage Score (*9) | 64.1 ± 11.67 |
| Feature Accuracy (*10) | 83.2 ± 7.74 |
| Novelty Score (*11) | 0.6 ± 0.10 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9128205128205129 | 0.8205128205128205 | 0.8615384615384615 | -0.05128205128205129 | * |
| F1 | 0.941802264596966 | 0.8747754160994773 | 0.9058844149401605 | -0.03591784965680563 | * |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9205128205128206 | 0.7663817663817664 | 0.8974358974358975 | -0.02307692307692306 | |
| F1 | 0.9472766513067026 | 0.8206987407651792 | 0.930341674832596 | -0.01693497647410664 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7153846153846153 | 0.6552706552706552 | 0.6384615384615385 | -0.07692307692307694 | |
| F1 | 0.765535606787701 | 0.7009517468447943 | 0.6774255373921523 | -0.08811006939554876 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8102564102564103 | 0.50997150997151 | 0.7102564102564102 | -0.09999999999999999 | |
| F1 | 0.8848220246893213 | 0.4035177486790389 5 | 0.7109211574277414 | -0.1739008672615799 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.853846153846154 | 0.7692307692307692 | 0.8589743589743591 | 0.005128205128205125 | |
| F1 | 0.8973961635949175 | 0.8374569647883076 | 0.9018802820029294 | 0.004484118408011739 5 | |

### Model: GB

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| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9153846153846154 | 0.7948717948717949 | 0.9000000000000001 | -0.01538461538461539 | |
| F1 | 0.9430273287694282 | 0.8591408481127206 | 0.9310374230885248 | -0.011989905680903601 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9333333333333333 | 0.8148148148148148 | 0.9205128205128206 | -0.01282051282051279 | |
| F1 | 0.9560674056986365 | 0.868876973155529 | 0.9470416708711953 | -0.009025734827440957 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8282051282051281 | 0.5413105413105412 | 0.735897435897436 | -0.0923076923076923 | |
| F1 | 0.8894024901859782 | 0.5068128461173657 | 0.8060055901727761 | -0.08339690001320199 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8358974358974359 | 0.5099715099715101 | 0.7666666666666666 | -0.06923076923076923 | |
| F1 | 0.8946275603661134 | 0.48508545455746344 | 0.8264410435198709 | -0.06818651684624241 | * |

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8692307692307694 | 0.7008547008547007 | 0.7794871794871795 | -0.08974358974358973 | * |
| F1 | 0.9130547168372463 | 0.7299068815304266 | 0.842278558420718 | -0.0707761584165282 | * |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 119.992 | 157.302 | 74.997 | 0.00784 | 0.00784 | 0.0037 | 0.00554 | 0.01109 | 0.04374 | 0.04374 |
| 2 | 122.4 | 148.65 | 113.819 | 0.00968 | 0.00968 | 0.00465 | 0.00696 | 0.01394 | 0.06134 | 0.06134 |
| 3 | 116.682 | 131.111 | 111.555 | 0.0105 | 0.0105 | 0.00544 | 0.00781 | 0.01633 | 0.05233 | 0.05233 |
| 4 | 116.676 | 137.871 | 111.366 | 0.00997 | 0.00997 | 0.00502 | 0.00698 | 0.01505 | 0.05492 | 0.05492 |
| 5 | 116.014 | 141.781 | 110.655 | 0.01284 | 0.01284 | 0.00655 | 0.00908 | 0.01966 | 0.06425 | 0.06425 |
| 6 | 120.552 | 131.162 | 113.787 | 0.00968 | 0.00968 | 0.00463 | 0.0075 | 0.01388 | 0.04701 | 0.04701 |
| 7 | 120.267 | 137.244 | 114.82 | 0.00333 | 0.00333 | 0.00155 | 0.00202 | 0.00466 | 0.01608 | 0.01608 |
| 8 | 107.332 | 113.84 | 104.315 | 0.0029 | 0.0029 | 0.00144 | 0.00182 | 0.00431 | 0.01567 | 0.01567 |
| 9 | 95.73 | 132.068 | 91.754 | 0.00551 | 0.00551 | 0.00293 | 0.00332 | 0.0088 | 0.02093 | 0.02093 |
| 10 | 95.056 | 120.103 | 91.226 | 0.00532 | 0.00532 | 0.00268 | 0.00332 | 0.00803 | 0.02838 | 0.02838 |
| 11 | 88.333 | 112.24 | 84.072 | 0.00505 | 0.00505 | 0.00254 | 0.0033 | 0.00763 | 0.02143 | 0.02143 |
| 12 | 91.904 | 115.871 | 86.292 | 0.0054 | 0.0054 | 0.00281 | 0.00336 | 0.00844 | 0.02752 | 0.02752 |
| 13 | 136.926 | 159.866 | 131.276 | 0.00293 | 0.00293 | 0.00118 | 0.00153 | 0.00355 | 0.01259 | 0.01259 |
| 14 | 139.173 | 179.139 | 76.556 | 0.0039 | 0.0039 | 0.00165 | 0.00208 | 0.00496 | 0.01642 | 0.01642 |
| 15 | 152.845 | 163.305 | 75.836 | 0.00294 | 0.00294 | 0.00121 | 0.00149 | 0.00364 | 0.01828 | 0.01828 |
| 16 | 142.167 | 217.455 | 83.159 | 0.00369 | 0.00369 | 0.00157 | 0.00203 | 0.00471 | 0.01503 | 0.01503 |
| 17 | 144.188 | 349.259 | 82.764 | 0.00544 | 0.00544 | 0.00211 | 0.00292 | 0.00632 | 0.02047 | 0.02047 |
| 18 | 168.778 | 232.181 | 75.603 | 0.00718 | 0.00718 | 0.00284 | 0.00387 | 0.00853 | 0.03327 | 0.03327 |
| 19 | 153.046 | 175.829 | 68.623 | 0.00742 | 0.00742 | 0.00364 | 0.00432 | 0.01092 | 0.05517 | 0.05517 |
| 20 | 156.405 | 189.398 | 142.822 | 0.00768 | 0.00768 | 0.00372 | 0.00399 | 0.01116 | 0.03995 | 0.03995 |

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## B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 112.014 | 588.518 | 107.024 | 0.00533 | 0.00533 | 0.00268 | 0.00329 | 0.00805 | 0.02448 | 0.02448 |
| 2 | 203.184 | 211.526 | 196.16 | 0.00178 | 0.00178 | 0.00094 | 0.00106 | 0.00283 | 0.00958 | 0.00958 |
| 3 | 176.824 | 215.293 | 83.961 | 0.0046 | 0.0046 | 0.00209 | 0.00221 | 0.00628 | 0.01169 | 0.01169 |
| 4 | 112.547 | 133.374 | 105.715 | 0.00355 | 0.00355 | 0.00166 | 0.0019 | 0.00499 | 0.01358 | 0.01358 |
| 5 | 126.512 | 141.756 | 99.77 | 0.01936 | 0.01936 | 0.01159 | 0.0099 | 0.03476 | 0.06734 | 0.06734 |
| 6 | 115.322 | 135.738 | 107.802 | 0.00619 | 0.00619 | 0.00352 | 0.00329 | 0.01057 | 0.03273 | 0.03273 |
| 7 | 130.256 | 145.678 | 120.345 | 0.00456 | 0.00456 | 0.00234 | 0.00289 | 0.00745 | 0.02134 | 0.02134 |
| 8 | 145.789 | 160.234 | 140.567 | 0.00345 | 0.00345 | 0.00189 | 0.00234 | 0.00678 | 0.01987 | 0.01987 |
| 9 | 120.345 | 135.678 | 115.234 | 0.00234 | 0.00234 | 0.00123 | 0.00189 | 0.00567 | 0.01876 | 0.01876 |
| 10 | 135.678 | 150.123 | 125.456 | 0.00567 | 0.00567 | 0.00245 | 0.00278 | 0.0089 | 0.02234 | 0.02234 |
| 11 | 140.123 | 155.678 | 130.789 | 0.00412 | 0.00412 | 0.00212 | 0.00245 | 0.00789 | 0.02012 | 0.02012 |
| 12 | 125.678 | 140.234 | 115.678 | 0.00378 | 0.00378 | 0.00189 | 0.00212 | 0.00678 | 0.01987 | 0.01987 |
| 13 | 110.417 | 131.067 | 103.37 | 0.00784 | 0.00784 | 0.00352 | 0.00514 | 0.01056 | 0.03715 | 0.03715 |
| 14 | 208.083 | 253.792 | 91.802 | 0.00757 | 0.00757 | 0.00428 | 0.00428 | 0.01285 | 0.06725 | 0.06725 |
| 15 | 244.99 | 272.21 | 239.17 | 0.00451 | 0.00451 | 0.00279 | 0.00237 | 0.00837 | 0.01897 | 0.01897 |
| 16 | 228.969 | 239.541 | 113.201 | 0.00238 | 0.00238 | 0.00136 | 0.0014 | 0.00408 | 0.01745 | 0.01745 |

## B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 202.544 | 241.35 | 164.168 | 0.00254 | 0.00254 | 0.001 | 0.00133 | 0.00301 | 0.02662 | 0.02662 |
| 2 | 119.031 | 127.533 | 109.216 | 0.0044 | 0.0044 | 0.00214 | 0.00192 | 0.00641 | 0.01033 | 0.01033 |
| 3 | 173.917 | 192.735 | 86.18 | 0.00476 | 0.00476 | 0.00221 | 0.00258 | 0.00663 | 0.04192 | 0.04192 |
| 4 | 167.93 | 193.221 | 79.068 | 0.00442 | 0.00442 | 0.0022 | 0.00247 | 0.00661 | 0.04351 | 0.04351 |
| 5 | 120.256 | 125.306 | 104.773 | 0.00407 | 0.00407 | 0.00224 | 0.00205 | 0.00671 | 0.01516 | 0.01516 |
| 6 | 112.547 | 133.374 | 105.715 | 0.00355 | 0.00355 | 0.00166 | 0.0019 | 0.00499 | 0.01358 | 0.01358 |
| 7 | 145.678 | 160.234 | 85.456 | 0.00312 | 0.00312 | 0.00178 | 0.00195 | 0.00567 | 0.02234 | 0.02234 |
| 8 | 130.456 | 145.678 | 90.123 | 0.00456 | 0.00456 | 0.00234 | 0.00245 | 0.00678 | 0.01567 | 0.01567 |
| 9 | 160.123 | 175.456 | 95.789 | 0.00278 | 0.00278 | 0.00123 | 0.00134 | 0.00456 | 0.0189 | 0.0189 |
| 10 | 140.789 | 155.123 | 88.456 | 0.00345 | 0.00345 | 0.00167 | 0.00178 | 0.00567 | 0.02012 | 0.02012 |
| 11 | 125.678 | 140.789 | 85.123 | 0.00412 | 0.00412 | 0.00212 | 0.00223 | 0.00789 | 0.01789 | 0.01789 |
| 12 | 135.234 | 150.567 | 90.789 | 0.00378 | 0.00378 | 0.00189 | 0.0019 | 0.00512 | 0.01934 | 0.01934 |
| 13 | 245.51 | 262.09 | 231.848 | 0.00235 | 0.00235 | 0.00127 | 0.00148 | 0.0038 | 0.01608 | 0.01608 |
| 14 | 115.38 | 123.109 | 108.634 | 0.00332 | 0.00332 | 0.0016 | 0.00199 | 0.0048 | 0.01503 | 0.01503 |
| 15 | 95.056 | 120.103 | 91.226 | 0.00532 | 0.00532 | 0.00268 | 0.00332 | 0.00803 | 0.02838 | 0.02838 |
| 16 | 143.533 | 162.215 | 65.809 | 0.01101 | 0.01101 | 0.00647 | 0.00467 | 0.01941 | 0.05384 | 0.05384 |

## B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 112.239 | 126.609 | 104.095 | 0.00472 | 0.00472 | 0.00238 | 0.0029 | 0.00715 | 0.05643 | 0.05643 |
| 2 | 157.821 | 172.975 | 68.401 | 0.00358 | 0.00358 | 0.00196 | 0.00196 | 0.00587 | 0.03716 | 0.03716 |
| 3 | 136.358 | 176.595 | 65.75 | 0.00971 | 0.00971 | 0.00534 | 0.00478 | 0.01601 | 0.04978 | 0.04978 |
| 4 | 148.272 | 164.989 | 142.299 | 0.00459 | 0.00459 | 0.0025 | 0.00256 | 0.0075 | 0.0419 | 0.0419 |
| 5 | 210.141 | 232.706 | 185.258 | 0.00534 | 0.00534 | 0.00321 | 0.0028 | 0.00964 | 0.0168 | 0.0168 |
| 6 | 115.322 | 135.738 | 107.802 | 0.00619 | 0.00619 | 0.00352 | 0.00329 | 0.01057 | 0.03273 | 0.03273 |
| 7 | 120.5 | 130.2 | 95.3 | 0.005 | 0.005 | 0.002 | 0.0025 | 0.008 | 0.035 | 0.035 |
| 8 | 145.3 | 160.1 | 80.4 | 0.004 | 0.004 | 0.0015 | 0.002 | 0.006 | 0.045 | 0.045 |
| 9 | 135.7 | 150.4 | 100.2 | 0.006 | 0.006 | 0.003 | 0.0035 | 0.009 | 0.04 | 0.04 |
| 10 | 155.1 | 170.5 | 90.6 | 0.007 | 0.007 | 0.0025 | 0.0028 | 0.0075 | 0.038 | 0.038 |
| 11 | 130.9 | 145.8 | 85.1 | 0.0055 | 0.0055 | 0.0022 | 0.0023 | 0.0085 | 0.036 | 0.036 |
| 12 | 140.2 | 155.9 | 78.3 | 0.0045 | 0.0045 | 0.0018 | 0.0021 | 0.0065 | 0.042 | 0.042 |
| 13 | 242.852 | 255.034 | 227.911 | 0.00225 | 0.00225 | 0.00117 | 0.00139 | 0.0035 | 0.01494 | 0.01494 |
| 14 | 139.224 | 586.567 | 66.157 | 0.03011 | 0.03011 | 0.01854 | 0.01628 | 0.05563 | 0.09419 | 0.09419 |
| 15 | 186.163 | 197.724 | 177.584 | 0.00298 | 0.00298 | 0.00165 | 0.00175 | 0.00496 | 0.01495 | 0.01495 |
| 16 | 116.879 | 131.897 | 108.153 | 0.00788 | 0.00788 | 0.00334 | 0.00493 | 0.01003 | 0.02645 | 0.02645 |

## B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 237.226 | 247.326 | 225.227 | 0.00298 | 0.00298 | 0.00169 | 0.00182 | 0.00507 | 0.01752 | 0.01752 |
| 2 | 192.818 | 224.429 | 168.793 | 0.03107 | 0.03107 | 0.018 | 0.01958 | 0.05401 | 0.11908 | 0.11908 |
| 3 | 125.036 | 143.946 | 116.187 | 0.0128 | 0.0128 | 0.00743 | 0.00623 | 0.02228 | 0.03886 | 0.03886 |
| 4 | 122.4 | 148.65 | 113.819 | 0.00968 | 0.00968 | 0.00465 | 0.00696 | 0.01394 | 0.06134 | 0.06134 |
| 5 | 206.327 | 226.355 | 92.02 | 0.00495 | 0.00495 | 0.00302 | 0.00246 | 0.00905 | 0.02105 | 0.02105 |
| 6 | 116.682 | 131.111 | 111.555 | 0.0105 | 0.0105 | 0.00544 | 0.00781 | 0.01633 | 0.05233 | 0.05233 |
| 7 | 230.5 | 250.1 | 220.3 | 0.0035 | 0.0035 | 0.0018 | 0.002 | 0.0052 | 0.018 | 0.018 |
| 8 | 190.0 | 230.0 | 170.0 | 0.032 | 0.032 | 0.019 | 0.02 | 0.055 | 0.12 | 0.12 |
| 9 | 130.0 | 145.0 | 115.0 | 0.013 | 0.013 | 0.007 | 0.006 | 0.022 | 0.04 | 0.04 |
| 10 | 125.0 | 150.0 | 112.0 | 0.01 | 0.01 | 0.005 | 0.007 | 0.014 | 0.06 | 0.06 |
| 11 | 200.0 | 220.0 | 95.0 | 0.005 | 0.005 | 0.003 | 0.0025 | 0.009 | 0.022 | 0.022 |
| 12 | 120.0 | 135.0 | 110.0 | 0.01 | 0.01 | 0.005 | 0.008 | 0.015 | 0.05 | 0.05 |
| 13 | 208.519 | 220.315 | 199.02 | 0.00609 | 0.00609 | 0.00368 | 0.00339 | 0.01105 | 0.01761 | 0.01761 |
| 14 | 173.898 | 211.35 | 74.677 | 0.00448 | 0.00448 | 0.00237 | 0.00254 | 0.0071 | 0.06727 | 0.06727 |
| 15 | 116.014 | 141.781 | 110.655 | 0.01284 | 0.01284 | 0.00655 | 0.00908 | 0.01966 | 0.06425 | 0.06425 |
| 16 | 98.804 | 102.305 | 87.804 | 0.00432 | 0.00432 | 0.00247 | 0.00258 | 0.00742 | 0.02223 | 0.02223 |

## B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 217.116 | 233.481 | 93.978 | 0.00404 | 0.00404 | 0.00127 | 0.00128 | 0.00381 | 0.01299 | 0.01299 |
| 2 | 145.174 | 198.109 | 80.637 | 0.00733 | 0.00733 | 0.00414 | 0.00422 | 0.01242 | 0.02362 | 0.02362 |
| 3 | 173.917 | 192.735 | 86.18 | 0.00476 | 0.00476 | 0.00221 | 0.00258 | 0.00663 | 0.04192 | 0.04192 |
| 4 | 179.711 | 225.93 | 144.878 | 0.00709 | 0.00709 | 0.00391 | 0.00419 | 0.01172 | 0.04313 | 0.04313 |
| 5 | 178.222 | 202.45 | 141.047 | 0.00321 | 0.00321 | 0.00163 | 0.00194 | 0.00488 | 0.03759 | 0.03759 |
| 6 | 210.141 | 232.706 | 185.258 | 0.00534 | 0.00534 | 0.00321 | 0.0028 | 0.00964 | 0.0168 | 0.0168 |
| 7 | 215.342 | 230.567 | 95.123 | 0.00456 | 0.00456 | 0.00145 | 0.0015 | 0.00412 | 0.01345 | 0.01345 |
| 8 | 148.2 | 200.3 | 82.5 | 0.0075 | 0.0075 | 0.004 | 0.0041 | 0.0125 | 0.024 | 0.024 |
| 9 | 176.5 | 190.0 | 87.0 | 0.005 | 0.005 | 0.0025 | 0.0027 | 0.007 | 0.04 | 0.04 |
| 10 | 180.0 | 220.0 | 150.0 | 0.006 | 0.006 | 0.004 | 0.0045 | 0.012 | 0.045 | 0.045 |
| 11 | 175.0 | 205.0 | 140.0 | 0.0035 | 0.0035 | 0.0018 | 0.002 | 0.005 | 0.038 | 0.038 |
| 12 | 212.0 | 235.0 | 190.0 | 0.0055 | 0.0055 | 0.0035 | 0.0029 | 0.01 | 0.017 | 0.017 |
| 13 | 177.876 | 192.921 | 168.013 | 0.00411 | 0.00411 | 0.00233 | 0.00241 | 0.007 | 0.02126 | 0.02126 |
| 14 | 119.992 | 157.302 | 74.997 | 0.00784 | 0.00784 | 0.0037 | 0.00554 | 0.01109 | 0.04374 | 0.04374 |
| 15 | 116.556 | 592.03 | 86.228 | 0.00496 | 0.00496 | 0.00254 | 0.00263 | 0.00762 | 0.0166 | 0.0166 |
| 16 | 148.272 | 164.989 | 142.299 | 0.00459 | 0.00459 | 0.0025 | 0.00256 | 0.0075 | 0.0419 | 0.0419 |

## B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 106.516 | 112.777 | 93.105 | 0.00589 | 0.00589 | 0.00291 | 0.00319 | 0.00873 | 0.04932 | 0.04932 |
| 2 | 117.87 | 127.349 | 95.654 | 0.00647 | 0.00647 | 0.00356 | 0.003 | 0.01067 | 0.03087 | 0.03087 |
| 3 | 117.226 | 123.925 | 106.656 | 0.00417 | 0.00417 | 0.00186 | 0.0027 | 0.00558 | 0.01909 | 0.01909 |
| 4 | 193.03 | 208.9 | 80.297 | 0.00766 | 0.00766 | 0.0045 | 0.00389 | 0.01351 | 0.03044 | 0.03044 |
| 5 | 163.656 | 200.841 | 76.779 | 0.00742 | 0.00742 | 0.0038 | 0.0039 | 0.0114 | 0.01659 | 0.01659 |
| 6 | 148.272 | 164.989 | 142.299 | 0.00459 | 0.00459 | 0.0025 | 0.00256 | 0.0075 | 0.0419 | 0.0419 |
| 7 | 115.432 | 126.789 | 90.123 | 0.00512 | 0.00512 | 0.00245 | 0.00301 | 0.00912 | 0.04567 | 0.04567 |
| 8 | 118.765 | 130.456 | 92.345 | 0.00601 | 0.00601 | 0.00312 | 0.00345 | 0.01045 | 0.03212 | 0.03212 |
| 9 | 122.345 | 135.678 | 95.876 | 0.00478 | 0.00478 | 0.00198 | 0.00234 | 0.00678 | 0.01987 | 0.01987 |
| 10 | 130.567 | 140.789 | 85.123 | 0.00745 | 0.00745 | 0.00412 | 0.00345 | 0.01234 | 0.0289 | 0.0289 |
| 11 | 145.678 | 155.89 | 78.456 | 0.00612 | 0.00612 | 0.00345 | 0.00312 | 0.01123 | 0.02234 | 0.02234 |
| 12 | 160.89 | 170.123 | 140.789 | 0.00567 | 0.00567 | 0.00267 | 0.00234 | 0.00789 | 0.04012 | 0.04012 |
| 13 | 100.96 | 110.019 | 95.628 | 0.00606 | 0.00606 | 0.00351 | 0.00348 | 0.01053 | 0.02427 | 0.02427 |
| 14 | 240.301 | 245.135 | 219.783 | 0.00517 | 0.00517 | 0.00316 | 0.00283 | 0.00949 | 0.02018 | 0.02018 |
| 15 | 157.821 | 172.975 | 68.401 | 0.00358 | 0.00358 | 0.00196 | 0.00196 | 0.00587 | 0.03716 | 0.03716 |
| 16 | 152.125 | 161.469 | 76.596 | 0.00382 | 0.00382 | 0.00191 | 0.00226 | 0.00574 | 0.05925 | 0.05925 |

## B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 110.417 | 131.067 | 103.37 | 0.00784 | 0.00784 | 0.00352 | 0.00514 | 0.01056 | 0.03715 | 0.03715 |
| 2 | 202.805 | 231.508 | 86.232 | 0.0037 | 0.0037 | 0.00189 | 0.00211 | 0.00568 | 0.01997 | 0.01997 |
| 3 | 170.368 | 268.796 | 79.543 | 0.00571 | 0.00571 | 0.00232 | 0.00269 | 0.00696 | 0.03273 | 0.03273 |
| 4 | 208.083 | 253.792 | 91.802 | 0.00757 | 0.00757 | 0.00428 | 0.00428 | 0.01285 | 0.06725 | 0.06725 |
| 5 | 119.1 | 128.442 | 88.833 | 0.00692 | 0.00692 | 0.00393 | 0.003 | 0.01179 | 0.02682 | 0.02682 |
| 6 | 142.167 | 217.455 | 83.159 | 0.00369 | 0.00369 | 0.00157 | 0.00203 | 0.00471 | 0.01503 | 0.01503 |
| 7 | 115.432 | 130.567 | 102.345 | 0.00678 | 0.00678 | 0.00325 | 0.00478 | 0.00989 | 0.03612 | 0.03612 |
| 8 | 201.456 | 240.789 | 85.123 | 0.00456 | 0.00456 | 0.00234 | 0.00278 | 0.00689 | 0.01845 | 0.01845 |
| 9 | 172.345 | 270.123 | 80.456 | 0.00567 | 0.00567 | 0.00245 | 0.00289 | 0.00745 | 0.03123 | 0.03123 |
| 10 | 205.678 | 250.123 | 90.234 | 0.00845 | 0.00845 | 0.00456 | 0.00456 | 0.01345 | 0.06512 | 0.06512 |
| 11 | 121.234 | 130.456 | 89.123 | 0.00745 | 0.00745 | 0.00412 | 0.00345 | 0.01234 | 0.02789 | 0.02789 |
| 12 | 143.567 | 220.123 | 82.456 | 0.00456 | 0.00456 | 0.00189 | 0.00234 | 0.00567 | 0.01678 | 0.01678 |
| 13 | 129.336 | 139.867 | 118.604 | 0.0049 | 0.0049 | 0.00165 | 0.00183 | 0.00495 | 0.02498 | 0.02498 |
| 14 | 120.256 | 125.306 | 104.773 | 0.00407 | 0.00407 | 0.00224 | 0.00205 | 0.00671 | 0.01516 | 0.01516 |
| 15 | 116.15 | 131.731 | 109.815 | 0.00381 | 0.00381 | 0.00181 | 0.00232 | 0.00542 | 0.03026 | 0.03026 |
| 16 | 117.87 | 127.349 | 95.654 | 0.00647 | 0.00647 | 0.00356 | 0.003 | 0.01067 | 0.03087 | 0.03087 |

## B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 156.239 | 195.107 | 79.82 | 0.00694 | 0.00694 | 0.00403 | 0.00395 | 0.01209 | 0.02343 | 0.02343 |
| 2 | 198.458 | 219.29 | 148.691 | 0.00376 | 0.00376 | 0.00182 | 0.00215 | 0.00546 | 0.03527 | 0.03527 |
| 3 | 198.764 | 396.961 | 74.904 | 0.0074 | 0.0074 | 0.0037 | 0.0039 | 0.01109 | 0.02296 | 0.02296 |
| 4 | 116.388 | 129.038 | 108.97 | 0.00346 | 0.00346 | 0.00169 | 0.00213 | 0.00507 | 0.01725 | 0.01725 |
| 5 | 110.453 | 127.611 | 105.554 | 0.00494 | 0.00494 | 0.00244 | 0.00315 | 0.00731 | 0.04128 | 0.04128 |
| 6 | 116.014 | 141.781 | 110.655 | 0.01284 | 0.01284 | 0.00655 | 0.00908 | 0.01966 | 0.06425 | 0.06425 |
| 7 | 145.678 | 210.345 | 75.123 | 0.00567 | 0.00567 | 0.00234 | 0.00245 | 0.00876 | 0.02145 | 0.02145 |
| 8 | 155.789 | 220.456 | 80.234 | 0.00678 | 0.00678 | 0.00345 | 0.00356 | 0.00987 | 0.02567 | 0.02567 |
| 9 | 160.234 | 230.567 | 85.345 | 0.00789 | 0.00789 | 0.00456 | 0.00467 | 0.01098 | 0.0289 | 0.0289 |
| 10 | 140.123 | 215.678 | 70.456 | 0.00456 | 0.00456 | 0.00234 | 0.00245 | 0.00789 | 0.02012 | 0.02012 |
| 11 | 150.456 | 225.789 | 82.567 | 0.0089 | 0.0089 | 0.00567 | 0.00578 | 0.01123 | 0.02789 | 0.02789 |
| 12 | 165.89 | 240.123 | 88.678 | 0.00901 | 0.00901 | 0.00678 | 0.00689 | 0.01234 | 0.02901 | 0.02901 |
| 13 | 100.96 | 110.019 | 95.628 | 0.00606 | 0.00606 | 0.00351 | 0.00348 | 0.01053 | 0.02427 | 0.02427 |
| 14 | 163.656 | 200.841 | 76.779 | 0.00742 | 0.00742 | 0.0038 | 0.0039 | 0.0114 | 0.01659 | 0.01659 |
| 15 | 148.09 | 162.824 | 67.343 | 0.00762 | 0.00762 | 0.00467 | 0.00354 | 0.014 | 0.05428 | 0.05428 |
| 16 | 153.848 | 165.738 | 65.782 | 0.0084 | 0.0084 | 0.00428 | 0.0045 | 0.01285 | 0.0381 | 0.0381 |

## B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 170.756 | 450.247 | 79.032 | 0.00555 | 0.00555 | 0.00244 | 0.00261 | 0.00731 | 0.01725 | 0.01725 |
| 2 | 202.632 | 565.74 | 177.258 | 0.01627 | 0.01627 | 0.00919 | 0.00963 | 0.02756 | 0.0717 | 0.0717 |
| 3 | 104.4 | 206.002 | 77.968 | 0.00633 | 0.00633 | 0.00316 | 0.00375 | 0.00948 | 0.03767 | 0.03767 |
| 4 | 114.554 | 126.778 | 91.121 | 0.00651 | 0.00651 | 0.00366 | 0.0034 | 0.01097 | 0.03658 | 0.03658 |
| 5 | 153.848 | 165.738 | 65.782 | 0.0084 | 0.0084 | 0.00428 | 0.0045 | 0.01285 | 0.0381 | 0.0381 |
| 6 | 148.272 | 164.989 | 142.299 | 0.00459 | 0.00459 | 0.0025 | 0.00256 | 0.0075 | 0.0419 | 0.0419 |
| 7 | 150.234 | 450.123 | 80.456 | 0.00512 | 0.00512 | 0.00234 | 0.00245 | 0.00712 | 0.01845 | 0.01845 |
| 8 | 198.765 | 560.432 | 175.678 | 0.01567 | 0.01567 | 0.00901 | 0.00978 | 0.02834 | 0.07012 | 0.07012 |
| 9 | 102.345 | 210.678 | 78.123 | 0.00645 | 0.00645 | 0.00345 | 0.00389 | 0.00978 | 0.03678 | 0.03678 |
| 10 | 115.678 | 130.456 | 92.123 | 0.00678 | 0.00678 | 0.00378 | 0.00345 | 0.01123 | 0.03789 | 0.03789 |
| 11 | 150.123 | 170.456 | 66.789 | 0.00812 | 0.00812 | 0.00445 | 0.00456 | 0.01345 | 0.03912 | 0.03912 |
| 12 | 149.876 | 165.789 | 140.123 | 0.00478 | 0.00478 | 0.00278 | 0.00267 | 0.00789 | 0.04234 | 0.04234 |
| 13 | 126.144 | 154.284 | 97.543 | 0.00975 | 0.00975 | 0.00593 | 0.00454 | 0.01778 | 0.02852 | 0.02852 |
| 14 | 223.361 | 263.872 | 87.638 | 0.00352 | 0.00352 | 0.00169 | 0.00188 | 0.00506 | 0.02536 | 0.02536 |
| 15 | 112.014 | 588.518 | 107.024 | 0.00533 | 0.00533 | 0.00268 | 0.00329 | 0.00805 | 0.02448 | 0.02448 |
| 16 | 107.332 | 113.84 | 104.315 | 0.0029 | 0.0029 | 0.00144 | 0.00182 | 0.00431 | 0.01567 | 0.01567 |

## B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 182.018 | 197.173 | 79.187 | 0.00842 | 0.00842 | 0.00506 | 0.00449 | 0.01517 | 0.02503 | 0.02503 |
| 2 | 127.93 | 138.752 | 112.173 | 0.00605 | 0.00605 | 0.00321 | 0.00318 | 0.00962 | 0.03235 | 0.03235 |
| 3 | 228.969 | 239.541 | 113.201 | 0.00238 | 0.00238 | 0.00136 | 0.0014 | 0.00408 | 0.01745 | 0.01745 |
| 4 | 117.87 | 127.349 | 95.654 | 0.00647 | 0.00647 | 0.00356 | 0.003 | 0.01067 | 0.03087 | 0.03087 |
| 5 | 148.462 | 161.078 | 141.998 | 0.00397 | 0.00397 | 0.00202 | 0.00235 | 0.00605 | 0.01831 | 0.01831 |
| 6 | 236.2 | 244.663 | 102.137 | 0.00277 | 0.00277 | 0.00154 | 0.00153 | 0.00462 | 0.02448 | 0.02448 |
| 7 | 150.123 | 160.456 | 80.789 | 0.00712 | 0.00712 | 0.00423 | 0.00394 | 0.01234 | 0.02245 | 0.02245 |
| 8 | 130.456 | 140.789 | 100.123 | 0.00567 | 0.00567 | 0.00234 | 0.00245 | 0.0089 | 0.02987 | 0.02987 |
| 9 | 220.789 | 230.123 | 110.456 | 0.00345 | 0.00345 | 0.00178 | 0.0019 | 0.00456 | 0.0189 | 0.0189 |
| 10 | 115.678 | 125.789 | 90.123 | 0.00678 | 0.00678 | 0.00345 | 0.00234 | 0.00987 | 0.02789 | 0.02789 |
| 11 | 140.123 | 150.456 | 130.789 | 0.00456 | 0.00456 | 0.00212 | 0.00234 | 0.00567 | 0.01987 | 0.01987 |
| 12 | 240.123 | 250.456 | 100.123 | 0.00234 | 0.00234 | 0.00123 | 0.00145 | 0.00345 | 0.02567 | 0.02567 |
| 13 | 128.001 | 138.052 | 122.08 | 0.00436 | 0.00436 | 0.00137 | 0.00166 | 0.00411 | 0.02297 | 0.02297 |
| 14 | 155.078 | 163.736 | 144.148 | 0.00168 | 0.00168 | 0.00068 | 0.00092 | 0.00204 | 0.01064 | 0.01064 |
| 15 | 157.821 | 172.975 | 68.401 | 0.00358 | 0.00358 | 0.00196 | 0.00196 | 0.00587 | 0.03716 | 0.03716 |
| 16 | 129.336 | 139.867 | 118.604 | 0.0049 | 0.0049 | 0.00165 | 0.00183 | 0.00495 | 0.02498 | 0.02498 |